

Geodesic Divergence Curvature, Torsional Decomposition, and the Commutator Bound

1 Kinematic Bound on Divergence Curvature

We study a curve $z(t)$ tracing a geodesic on a Hessian manifold defined by a self-concordant barrier ϕ .

Theorem 1. *Let $z(t)$ be a geodesic parameterized by arc length with constant kinetic energy $E = \|\dot{z}\|_{H(z)}^2$. Let z_* be an arbitrary target point. Then the second time derivative of the generalized Jeffreys divergence $V(z) = \langle z - z_*, \nabla\phi(z) - \nabla\phi(z_*) \rangle$ is given by:*

$$\ddot{V} = 2E - \langle \ddot{z}, U \rangle_z \quad (1)$$

where $U = (z - z_*) - H(z)^{-1}(\nabla\phi(z) - \nabla\phi(z_*))$ is the exact aggregated residual vector, and the inner product is in the local metric $\langle a, b \rangle_z = a^T H(z) b$.

Proof. Taking the first time derivative of $V(z)$ along the geodesic $z(t)$:

$$\dot{V} = \langle \dot{z}, \nabla\phi(z) - \nabla\phi(z_*) \rangle + \langle z - z_*, H(z)\dot{z} \rangle \quad (2)$$

Taking the second time derivative yields:

$$\begin{aligned} \ddot{V} = & \langle \ddot{z}, \nabla\phi(z) - \nabla\phi(z_*) \rangle + \langle \dot{z}, H(z)\dot{z} \rangle \\ & + \langle \dot{z}, H(z)\dot{z} \rangle + \langle z - z_*, \dot{H}(z)\dot{z} \rangle + \langle z - z_*, H(z)\ddot{z} \rangle \end{aligned} \quad (3)$$

Recognizing that the kinetic energy is $E = \langle \dot{z}, H(z)\dot{z} \rangle$, we substitute $2E$ and group the acceleration terms:

$$\ddot{V} = 2E + \langle \ddot{z}, \nabla\phi(z) - \nabla\phi(z_*) + H(z)(z - z_*) \rangle + D^3\phi(z)[\dot{z}, \dot{z}, z - z_*] \quad (4)$$

By the definition of a geodesic on a Hessian manifold, the acceleration $\ddot{z}$ is driven by the Levi-Civita connection:

$$\ddot{z} = -\frac{1}{2}H(z)^{-1}D^3\phi(z)[\dot{z}, \dot{z}, \cdot] \quad (5)$$

Applying this identity to the vector $w = z - z_*$ rewrites the third derivative term as a local inner product with the acceleration:

$$D^3\phi(z)[\dot{z}, \dot{z}, z - z_*] = -2\langle \ddot{z}, H(z)(z - z_*) \rangle \quad (6)$$

Substituting this back into Equation 4 results in the exact cancellation of one of the primal residual terms:

$$\begin{aligned}\ddot{V} &= 2E + \langle \ddot{z}, \nabla\phi(z) - \nabla\phi(z_*) + H(z)(z - z_*) \rangle - 2\langle \ddot{z}, H(z)(z - z_*) \rangle \\ &= 2E + \langle \ddot{z}, \nabla\phi(z) - \nabla\phi(z_*) - H(z)(z - z_*) \rangle\end{aligned}\tag{7}$$

Factoring out $-H(z)$ reveals the exact generalized aggregated residual U :

$$\begin{aligned}\ddot{V} &= 2E - \langle \ddot{z}, H(z) \left[(z - z_*) - H(z)^{-1}(\nabla\phi(z) - \nabla\phi(z_*)) \right] \rangle \\ &= 2E - \langle \ddot{z}, U \rangle_z\end{aligned}\tag{8}$$

□

2 The Commutator and Torsional Decomposition

For a general self-concordant barrier $\phi(z)$, the metric-warping operator $M_u(v) = H(z)^{-1}D^3\phi(z)[u, v, \cdot]$ equips the tangent space with a commutative (but generally non-associative) algebraic product $u \bullet v = M_u(v)$. The local geometric square is $u^2 = M_u(u)$.

We formally define the **Sectional Curvature Commutator** as the Lie bracket of the multiplication operators:

$$\mathcal{C}(u) = [M_u, M_{u^2}] = M_u M_{u^2} - M_{u^2} M_u\tag{9}$$

By expanding the generalized residual U via the operator algebra along the step $\Delta = z_* - z$, the residual splits strictly into a power-associative component U_{sym} and a non-associative torsional component U_{torsion} :

$$U = U_{\text{sym}} + U_{\text{torsion}}\tag{10}$$

The decomposition of the inner product yields:

$$\langle \ddot{z}, U \rangle_z = \langle \ddot{z}, U_{\text{sym}} \rangle_z + \langle \ddot{z}, \mathcal{C}(\Delta)[v] \rangle_z\tag{11}$$

where v encapsulates higher-order remainders. The energy leakage in the inner product exists purely because $\mathcal{C}(\Delta) \neq 0$.

3 Recovering the Symmetric Cone Case and the Exact Formula

If the barrier possesses the self-scaled symmetry $\phi(-\nabla\phi(z)) = \phi(z) + c$, the operators form a Euclidean Jordan Algebra. The commutator universally vanishes ($\mathcal{C} \equiv 0 \implies U_{\text{torsion}} = 0$), so $U = U_{\text{sym}}$.

In this perfect geometry, we map the residual to an isotropic frame using the quadratic representation $P(z) = H(z)^{-1}$. Let the relative stretch vector be $w = P(z)^{-1/2}z_*$. The primal acceleration maps to $\tilde{z} = -\frac{1}{2}\tilde{z}^2$.

The dual-gradient mapping acts as the Jordan inverse. The generalized residual $U = (z - z_*) - P(z)(-\nabla\phi(z) + \nabla\phi(z_*))$ maps identically to the isotropic frame as $\tilde{U}_{sym} = \mathbf{21} - w - w^{-1}$. Thus, the inner product evaluates to the spectacularly clean associative formula:

$$\langle \ddot{z}, U_{sym} \rangle_z = -\frac{1}{2} \langle \ddot{z}^2, \mathbf{21} - w - w^{-1} \rangle \quad (12)$$

Since $w \approx \mathbf{1} + \Delta w$, a Taylor expansion yields $\mathbf{21} - w - w^{-1} \approx -(\Delta w)^2$. The symmetric residual flawlessly produces the pure quadratic centering force without higher-order torsion.

4 Polyhedral Cone

Theorem 2 (Polyhedral Commutator and Inner Product Decomposition). *Let $\mathcal{M} = \{z \in \mathbb{R}^n \mid Az > 0\}$ be a polyhedral cone equipped with the logarithmic barrier $F(z) = -\sum_{i=1}^m \ln(a_i^T z)$. Let $H(z) = \nabla^2 F(z)$ be the local metric. For a tangent vector u , let $\tilde{u} = S^{-1}Au$ be its lift into the scaled slack space, where $S = \text{diag}(Az)$. Let $P = S^{-1}A(A^T S^{-2}A)^{-1}A^T S^{-1}$ be the orthogonal projection matrix onto the column space of $S^{-1}A$.*

Then, the following hold:

1. **(Commutator Formula):** *The sectional curvature commutator $\mathcal{C}(u) = [M_u, M_{u^2}]$, lifted to the slack space, evaluates exactly to:*

$$\tilde{\mathcal{C}}(\tilde{u})[\tilde{v}] = P(\tilde{u} \circ P(P(\tilde{u}^{\circ 2}) \circ \tilde{v})) - P(P(\tilde{u}^{\circ 2}) \circ P(\tilde{u} \circ \tilde{v})) \quad (13)$$

where $\circ$ denotes the element-wise Hadamard product.

2. **(Inner Product Decomposition):** *For a continuous geodesic $z(t)$ and a target point z_* , define the relative stretch vector $w = S^{-1}Az_*$. The inner product between the acceleration $\ddot{z}$ and the exact aggregated residual $U = (z - z_*) - H(z)^{-1}(\nabla F(z) - \nabla F(z_*))$ evaluates exactly to:*

$$\langle \ddot{z}, U \rangle_z = (\ddot{z}^{\circ 2})^T P(\mathbf{21} - w - w^{-1}) \quad (14)$$

Proof. Part 1: The Commutator Formula.

The local metric is $H(z) = A^T S^{-2}A$. The third derivative tensor evaluated in directions u and v is $D^3 F(z)[u, v, \cdot] = -2A^T S^{-3} \text{diag}(Au)Av$. The metric-warping operator is defined as $M_u(v) = H(z)^{-1}D^3 F(z)[u, v, \cdot]$. Dropping the -2 scale factor for the structural commutator mapping, we map the operator into the slack space via $\tilde{M}_{\tilde{u}}(\tilde{v}) = S^{-1}AM_u(v)$. Substituting $H(z)^{-1}$ gives:

$$\begin{aligned} \tilde{M}_{\tilde{u}}(\tilde{v}) &= \left(S^{-1}A(A^T S^{-2}A)^{-1}A^T S^{-1} \right) (S^{-1}Au \circ S^{-1}Av) \\ &= P(\tilde{u} \circ \tilde{v}) \end{aligned}$$

The local geometric square lifts to $\tilde{u}_M^{\circ 2} = P(\tilde{u} \circ \tilde{u}) = P(\tilde{u}^{\circ 2})$. Substituting these into the Lie bracket $\mathcal{C}(u)[v] = M_u(M_{u^2}(v)) - M_{u^2}(M_u(v))$ directly yields the lifted commutator formula:

$$\tilde{\mathcal{C}}(\tilde{u})[\tilde{v}] = P(\tilde{u} \circ P(P(\tilde{u}^{\circ 2}) \circ \tilde{v})) - P(P(\tilde{u}^{\circ 2}) \circ P(\tilde{u} \circ \tilde{v}))$$

Part 2: The Inner Product Decomposition.

We first lift the exact aggregated residual U into the slack space. We have $\nabla F(z) = -A^T S^{-1} \mathbf{1}$ and $\nabla F(z_*) = -A^T S_*^{-1} \mathbf{1}$. By definition $w = S^{-1} s_*$, so $S_*^{-1} = S^{-1} \text{diag}(w^{-1})$. Therefore:

$$\nabla F(z) - \nabla F(z_*) = -A^T S^{-1} (\mathbf{1} - w^{-1})$$

Applying the inverse Hessian and projecting into the slack space via $S^{-1} A$ yields $-P(\mathbf{1} - w^{-1})$. The primal difference lifts as $S^{-1} A(z - z_*) = \mathbf{1} - w$. The lifted residual is thus exactly:

$$\tilde{U} = (\mathbf{1} - w) + P(\mathbf{1} - w^{-1})$$

Next, from the Levi-Civita connection $\ddot{z} = -\frac{1}{2} H(z)^{-1} D^3 F(z) [\dot{z}, \dot{z}, \cdot]$, substituting the 3rd derivative perfectly cancels the constants, yielding the lifted acceleration $\tilde{\ddot{z}} = P(\ddot{z}^{\circ 2})$. The local inner product is the standard Euclidean product in the lifted space:

$$\langle \ddot{z}, U \rangle_z = \tilde{\ddot{z}}^T \tilde{U} = P(\tilde{\ddot{z}}^{\circ 2})^T \left((\mathbf{1} - w) + P(\mathbf{1} - w^{-1}) \right)$$

Since P is a symmetric orthogonal projection matrix ($P^T = P = P^2$), and the stretch vector w strictly resides in the column space ($P(w) = w$, and assuming $P(\mathbf{1}) = \mathbf{1}$), the projection distributes smoothly:

$$\begin{aligned} \langle \ddot{z}, U \rangle_z &= (\tilde{\ddot{z}}^{\circ 2})^T P(\mathbf{1} - w + \mathbf{1} - w^{-1}) \\ &= (\tilde{\ddot{z}}^{\circ 2})^T P(2\mathbf{1} - w - w^{-1}) \end{aligned}$$

This completes the proof. □